

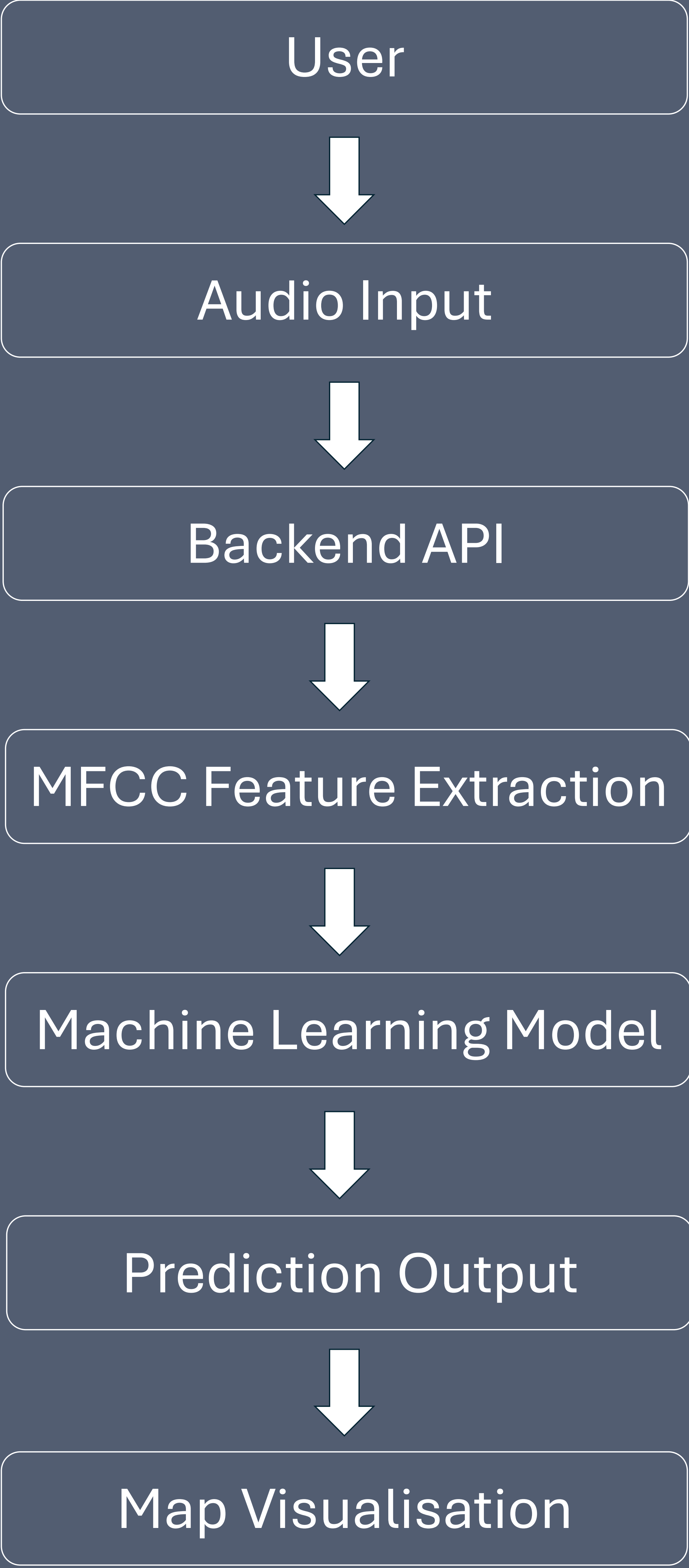
Mapping the Spoken Word:

A Geo-Spatial Machine Learning Approach to Irish Accent Recognition

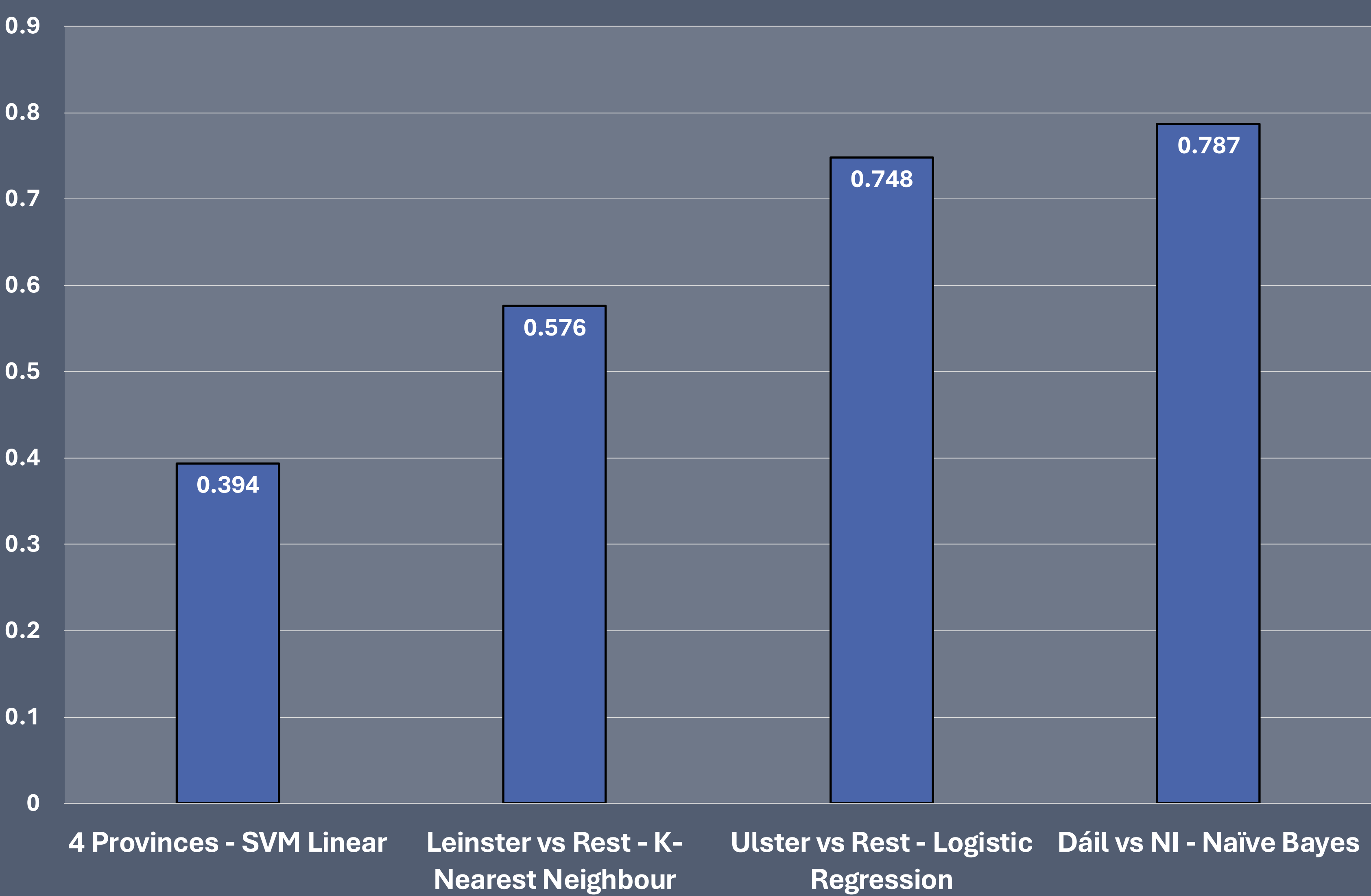
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System Diagram



Balanced Accuracy of Different Models on Different Data



Introduction

- The Irish accent varies massively across the little island of Ireland and is a major part of the Irish identity.
- Using machine learning we can analyse and classify speech patterns ie. Accents.
- My project explores a system for predicting and mapping Irish accents from self-created datasets.

Project Aim

To design and implement a machine learning system for Irish accent classification and geospatial visualisation, supported by an experimental evaluation of model performance.

Key Findings

- Province-level classification performed poorly (~0.39 balanced accuracy).
- Binary classification tasks achieved significantly higher performance (~0.75–0.79).
- Accent differences are more detectable at broader regional levels.
- Model performance is strongly influenced by dataset quality and balance.

